K-INJECTIVE COMPLEXES INVERT QUASI-ISOMORPHISMS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- K-injective complexes. lemma-K-injective (same chapter, carded in the old generation). Supplies the defining lifting property of K-injective complexes.
- What the localization inverts. lemma-derived-inverts (same chapter, cited in-span; not carded). Identifies quasi-isomorphisms as the inverted class.

Source reference. Original lemma 070I: sources/stacks-derived/pinned-original-source.tex, source lines 9849–9849. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SB: sources/stacks-derived/pinned-original-source.tex, source lines 4855–4855. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $F : K(\mathcal{A}) \to \mathcal{D}'$ be an exact functor of triangulated categories. Then RF is defined at every complex in $K(\mathcal{A})$ which is quasi-isomorphic to a K-injective complex. In fact, every K-injective complex computes RF.*

Proof. By Lemma 05SB it suffices to show that RF is defined at a K-injective complex, i.e., it suffices to show a K-injective complex $I^\bullet$ computes RF. Consider a quasi-isomorphism $I^\bullet \to N^\bullet$. By Lemma 070I it has a left inverse. Thus $I^\bullet \to I^\bullet$ is a final object of $I^\bullet/\mathrm{Qis}(\mathcal{A})$ and we win. □

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